

Research Hackathon Problem Statement

Theme: Aerospace, Space Science & Engineering **Duration:** 24 Hours **Institution:** IIT Kanpur - Institute Research Symposium 2025 **Submitted by:** Dr. Soumyabrata Chakrabarty, Professor of Practice **Department:** Space, Planetary & Astronomical Sciences & Engineering (SPASE), IIT Kanpur

Title

Mission Concept for Mitigating the Growing Threat of Space Debris in Low Earth Orbit (LEO)

1. Context / Background

The Low Earth Orbit (LEO) region (altitude 200-2000 km) has become increasingly congested with operational satellites, defunct spacecraft, rocket bodies, and fragments resulting from collisions and explosions.

With the exponential rise of mega-constellations such as Starlink, OneWeb, and Kuiper, the risk of collisions and cascading fragmentation (Kessler Syndrome) is becoming a critical concern for the long-term sustainability of space operations.

Research in this evolving field is crucial in space debris management because it provides the scientific understanding, technological innovation, and policy frameworks needed to ensure the long-term sustainability of space activities.

2. Objective

Conceptualize, design, and present a feasible and innovative solution that contributes to debris monitoring, mitigation, and removal.

Consider a **non-cooperative small satellite (roughly 500mm cuboid)** as debris target. Your solution should take care:

- The chaser spacecraft should not get damaged
- The target also should not get further fragmented

3. Key Tasks

List 3-5 major tasks or deliverables participants must complete.

Task No.	Description	Expected Output
1	Detection & Tracking	Innovative methods detecting space debris.Use of AI/ML or data fusion for improving orbit prediction and collision avoidance
2	Mitigation Strategies	Design improvements for spacecraft end-of-life disposal.Passive or active deorbiting techniques
3	Debris Removal Concepts	Novel mission concept for active debris removal (ADR).Your concept for capture mechanisms.
4	Summarize Insights	Short report / presentation

4. Expected Deliverables

Clearly specify what teams must submit at the end:

- **Problem Definition** – Identify the key challenge and its relevance.
- **Proposed Approach/Concept** – Describe your solution clearly.
- **System Design or Model** – Include diagrams, simulations/animations, or flowcharts as applicable
- **Feasibility Assessment** – Discuss potential advantages, limitations, and implementation roadmap.
- **Innovation & Impact** – Explain how your idea improves space sustainability.

5. Judging Criteria

Criterion	Weight	Description
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Innovation & Originality	25%	Rigour, clarity, originality
Technical Feasibility	25%	Tracking and Detection Feasibility, Active Debris Removal (ADR) Feasibility
Clarity & Quality of Presentation	20%	Visuals, clarity, discussion, creativity
Impact on Space Sustainability	20%	Safe, accessible, and usable space particle environment
Teamwork & Interdisciplinary Approach	10%	Collaboration Between Technical Disciplines

6. Resources (Optional)

- A. Ledkov, V. Aslanov, Review of contact and contactless active space debris removal approaches, Prog. Aerosp. Sci. 134 (2022) 100858.
- M. Shan, J. Guo, E. Gill, Review and comparison of active space debris capture and removal methods, Prog. Aero. Sci. 80 (2016) 18-32.

7. Suggested Reading / Keywords

Keywords: Space Debris / Orbital Debris, Low Earth Orbit (LEO), Space Sustainability, Debris Mitigation, Active Debris Removal (ADR), Collision Avoidance, Tracking and Monitoring, Space Situational Awareness (SSA), Orbital Dynamics, Kessler Syndrome, End-of-Life Disposal, Passivation, Deorbit Mechanism / Drag Sail / Tether, Laser Ablation / Robotic Capture, Risk Assessment, Space Traffic Management

Submission Format

- **File Name:** HackathonPS___.pdf
- **Submit via:** In form of poster/presentation